

Lab 5: Introduction to FANUC Hardware

EE0849 Introduction to Robotics

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IMPORTANT — Read Before You Begin

This is your first lab with actual hardware. Before touching anything:

1. Listen to the complete safety briefing
2. Know the location of ALL emergency stops
3. **NEVER** work alone with the robot

How This Lab Works

The instructor demonstrates each operation while the group observes. Selected students take turns at the teach pendant to perform the key steps (jogging, teaching positions, running programs). Follow along and fill in the tables as we go.

1 Part 1: Safety & Robot Overview

1.1 Emergency Stops

E-Stop Locations

As the instructor demonstrates, identify each device and record its location.

Device	Location	Action
E-Stop Button (Controller Panel)		
E-Stop Button (Teach Pendant)		
Safety Gate Switch		
Deadman Switch (Pendant, side grip)		

1.2 Operating Modes

Mode	Indicator	Who Can Move Robot	Safety
AUTO	Green	Controller only	Gate must be closed
T1 (Teach)	White	Pendant (slow, ≤ 250 mm/s)	Deadman required
T2	–	Pendant (full speed)	Deadman required

For this lab, we use T1 mode ONLY.

1.3 Robot Components

As the instructor points out each component, note its location:

Component	Function
Base (J1)	
Shoulder (J2)	
Elbow (J3)	
Wrist Roll / Pitch / Yaw (J4–J6)	
Tool Flange + End Effector	

1.4 Safety Acknowledgment

Safety Checklist

- ☐ I know all emergency stop locations
- ☐ I understand the deadman switch (hold — don't squeeze or release)
- ☐ I will never enter the cell in AUTO mode
- ☐ I will ask before any new operation

Signature: _____

Date: _____

2 Part 2: Teach Pendant & Jogging

2.1 Pendant Controls

Exercise: Identify Pendant Controls

Locate each control on the teach pendant as the instructor walks through them.

Control	Function
Deadman Switch (side)	Hold to enable motion
SHIFT + Jog Keys (+/-)	Move selected axis
COORD Key	Switch between JOINT / WORLD / TOOL
Speed Override	Adjust motion speed (%)
MENU / SELECT	Access menus / choose programs
FWD / BWD	Step through program lines

2.2 Jogging in Joint Mode

Exercise: Joint Jogging

1. Press **COORD** until display shows “JOINT”
2. Hold deadman + **SHIFT** + press jog keys for each joint
3. Record the direction of motion for each joint

Joint	+ Direction	- Direction
J1		
J2		
J3		
J4		
J5		
J6		

2.3 Jogging in World Mode

Exercise: World Jogging

Switch to WORLD mode and jog along X, Y, Z. Observe how the robot moves differently than in JOINT mode.

Joint vs World Jogging

Joint: Each motor moves independently — fast repositioning, curved path.

World: End-effector moves in straight Cartesian lines — slower, predictable path.

3 Part 3: Teaching Positions & Writing a Program

3.1 Teach Positions

What Is a Taught Position?

A taught position stores joint angles (J1–J6) **or** a Cartesian pose (X, Y, Z, W, P, R) plus configuration flags.

Exercise: Teach 5 Positions

1. Jog to a safe home position → record as **P[1]**
 2. Jog above the pick point (approach) → record as **P[2]**
 3. Jog down to the pick point → record as **P[3]**
 4. Jog above the place point → record as **P[4]**
 5. Jog down to the place point → record as **P[5]**
- To record: **MENU** → POSITION → Position Reg → select slot → **F2 [RECORD]**

Record your positions (joint values):

Position	Purpose	J1	J2	J3	J4	J5	J6
P[1]	Home						
P[2]	Pick approach						
P[3]	Pick						
P[4]	Place approach						
P[5]	Place						

3.2 Write a Pick-and-Place Program

Exercise: Create and Test a Program

Create a new program called LAB5_[YourInitials] with the following structure. Run it in **single-step mode** first (FWD one line at a time), then continuously.

```

/PROG PICK_PLACE
/MN
  1: J P[1] 100% FINE      ; Home
  2: J P[2] 50% FINE      ; Approach pick
  3: L P[3] 100mm/sec FINE ; Down to pick
  4: PAUSE                ; (Gripper close)
  5: L P[2] 100mm/sec FINE ; Retract
  6: J P[4] 50% FINE      ; Approach place
  7: L P[5] 100mm/sec FINE ; Down to place
  8: PAUSE                ; (Gripper open)
  9: L P[4] 100mm/sec FINE ; Retract
 10: J P[1] 100% FINE     ; Return home
/END

```

Motion Commands

J = Joint motion (curved path, fast). **L** = Linear motion (straight path, Cartesian).
FINE = stop at the point. **CNTn** = blend through (0 = tight, 100 = wide arc).

4 Part 4: Motion Comparison & Wrap-Up

4.1 Joint vs Linear Motion

Exercise: Compare J and L Motion

Run two short programs that move between the same two positions. Observe the difference in path and speed.

Program A: J P[1] 50% FINE then J P[5] 50% FINE

Program B: J P[1] 50% FINE then L P[5] 200mm/sec FINE

Aspect	Joint (J) Motion	Linear (L) Motion
Path shape		
Speed consistency		
When to use		

4.2 Questions

1. When would you prefer Joint motion over Linear?
2. What happens if Linear motion passes through a singularity?
3. How would you modify the program to pick from multiple locations?

4.3 Lab Completion

- ☐ Completed safety acknowledgment (Part 1)
- ☐ Jogged in Joint and World modes (Part 2)
- ☐ Taught 5 positions and created a pick-place program (Part 3)
- ☐ Compared Joint vs Linear motion (Part 4)

Instructor Signature: _____

Date: _____